

Sowmya Sree Challapalli

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EDUCATION

2023 - present	Bachelor of Technology at SRM University	(CGPA: 9.13)
2021 - 2023	Class 12th SSC Board at NRI Junior College	(Percentage: 97.5%)
2021	Class 10th SSC Board at Sri Chaitanya High School	(Percentage: 98.6%)

SKILLS

Programming Languages	Python, Java
Machine Learning & AI	XGBoost, OpenCV, Transformers, Model Evaluation, Pandas, NumPy, Matplotlib, Feature Engineering, Data Visualization
Database	SQL, PostgreSQL

WORK EXPERIENCE

Research Intern	Feb 2025 - present
• Comparative Analysis of Classical and Hybrid Quantum Recurrent Models for Photovoltaic Power Forecasting → Developed an HQLSTM model with variational quantum circuits for short-term PV forecasting, achieving RMSE (0.0693) and R^2 (0.9074). • Comparative Study of HQNN and MLP Models for Photovoltaic Power Forecasting → Proposed an HQNN framework with statistical features and temporal embeddings, achieving R^2 (0.9214) and RMSE (0.0558).	

PROJECTS

• Academic Result Management System [Link] → Developed a secure, database-driven web application to manage, update, and publish students' academic results using HTML, CSS, JavaScript, PHP, and MySQL. → Designed a structured ER model and normalized relational database schema to efficiently organize student, subject, branch, semester, and result data. • SmartChat AI [Link] → Built a full-stack AI chat app using FastAPI, React, and PostgreSQL. → Integrated Ollama with Gemma 3:3B model for real-time interaction with local LLMs. → Implemented authentication and chat history with a responsive UI.

CERTIFICATIONS

• Java SE 17 Developer Certification (<i>Oracle</i>) [Link] → Built structured and efficient programs focusing on application logic and error handling. <i>Skills:</i> OOP , Multithreading, File I/O, JVM Concepts
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